

ABDUL HANAN

Full Stack AI Engineer

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SUMMARY

Full Stack AI Engineer with end-to-end product delivery experience spanning React frontends, FastAPI backends, LLM/RAG integrations, and cloud-native AWS deployments. Skilled in building AI-powered SaaS applications—from vector database architecture and retrieval pipelines to containerized production systems with clean API design and seamless frontend-backend integration.

EDUCATION

FAST National University of Computer & Emerging Sciences

B.S. in Artificial Intelligence | August 2022 – May 2026

EXPERIENCE

Full Stack AI Engineer — Contract

Cognara Training | November 2025 – April 2026 | React, FastAPI, LangChain, pgvector, Docker, AWS

- Led end-to-end delivery of APD Listener Tool and Tradie Migration Platform—full-stack AI products from architecture through cloud deployment.
- Built React + FastAPI stack with JWT RBAC auth, PostgreSQL/pgvector data layer, LangChain RAG for AI document search, and Docker deployments on AWS EC2/Nginx.
- Integrated OpenAI APIs for intelligent visa case matching; AWS S3 + multithreaded queues for high-throughput audio-to-text pipelines.
- Taught international students Python from beginner to advanced level, covering OOP, REST APIs, web scraping, automation, backend development, and AI/ML integration patterns.

AI Engineer Intern

One Tech and AI | July 2025 – September 2025 | RAG, LLMs, Docker, FastAPI

- Built and shipped RAG-based chatbot apps and AI agent systems with production API layers, managing full retrieval pipeline orchestration and Docker containerization.
- Participated in Quantum AI research exploring practical hybrid quantum-classical ML applications.

SKILLS & TOOLS

Frontend: React, Vercel, REST API integration, real-time UI, file upload/download

Backend & APIs: Python, FastAPI, PostgreSQL, Neo4j, JWT/RBAC, SMTP workflows, SQL

AI & ML: LangChain, OpenAI API, RAG pipelines, pgvector, FAISS, Hugging Face, CLIP, MiniLM, PyTorch

Cloud & DevOps: AWS (EC2, S3, Lambda, SageMaker, IAM), Docker, Nginx, GitHub Actions, MLflow, Airflow

SELECTED PROJECTS

Tradie Migration Platform — AI SaaS Application

React, FastAPI, PostgreSQL, LangChain, OpenAI, pgvector, Docker, AWS | Mar – Apr 2026

- Full-stack AI SaaS for overseas tradesperson migration: React frontend, FastAPI backend, 6-role RBAC/JWT auth, LangChain+pgvector RAG for intelligent document search and visa case management.
- Integrated OpenAI embeddings for semantic candidate matching; automated Gmail SMTP/OTP flows; Docker-deployed on AWS EC2/Nginx.

APD Listener Tool — Full Stack Audio AI Platform

React, FastAPI, Docker, AWS EC2/S3, Nginx, Vercel | Nov 2025 – Feb 2026

- Commercial audio intelligence platform: React/Vercel frontend + FastAPI/EC2 backend, AWS S3 media delivery, multithreaded queue processing for concurrent high-throughput audio conversion.

Doctor Path AI — Clinical Decision Support

pgvector, Neo4j, FAISS, Docker, FastAPI, MedGemma | Aug 2025 – Feb 2026

- Full-stack clinical AI platform on MIMIC-III: multimodal patient record processing, hybrid retrieval (pgvector + Neo4j + FAISS), treatment recommendation APIs via FastAPI.

Hybrid E-Commerce Recommendation Engine

FAISS, Annoy, CLIP, PyTorch, MiniLM | Mar – May 2025

- Multimodal AI recommendation system fusing CLIP image + MiniLM text embeddings; two-stage Annoy→FAISS retrieval balancing speed and recommendation accuracy at scale.

HONORS & ACTIVITIES

- Top 50 Nationally — ICPC 2026 | Head of Job Fair 2026, Co-Head 2025, Coordinator 2024 — FAST NU
- Teaching Assistant: Generative AI, Deep Learning, DB Systems, Assembly Language — FAST NU CFD | 350+ LeetCode
- Python Instructor — Taught international students Python (beginner to advanced): OOP, automation, REST APIs, web scraping, backend development, and AI/ML concepts | July 2025 – January 2026
- Level 2 Fiverr Freelancer — AI Automation & Full Stack AI Development